

Akul Ramasubban

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Education

Purdue University

B.S in Mechanical Engineering- Presidential Scholar

West Lafayette, IN

May 2027

Technical Experience

Tesla

Cell Manufacturing Equipment Design/Process Engineering Intern

January 2026 - August 2026

- Improved equipment availability ~**10%** by building a custom accelerated-test jig to match vibration frequency of equipment to simulate long-term failure of fastening mechanism and reduce future quality equipment shutdowns.
- Engineered and deployed an autonomous production tool with timed actuation to correct misaligned current collectors from inconsistent supplier process, eliminating the floor's largest availability/IEE loss and scrap.
- Conducted RCA on cell ultrasonic welding process suffering significant yield loss; applied DOE and JMP-driven insights to implement production changes that improved downstream cell yield by ~**10,000** cells/week.
- Designed and implemented an equipment upgrade that sharply reduced cell contamination, while simultaneously reducing maintenance downtime by **25%**, helping increase cell production targets to support cybercab production.
- Created and deployed a protective barrier that prevented equipment failure on production equipment using production trials and rapid prototyping, leading to an availability improvement of **> 2.5%** and **\$1000** per failure.
- Led a fleetwide effort to standardize ultrasonic weld power and performed recalibration on each tool to reduce the occurrence of asymmetric weld strength by **20%** on average, improving equipment reliability and consistency.

Ford Motor Company

Product Development Intern- Forward Model

May 2025 - August 2025

- Facilitated a Six Sigma 3L5Y root-cause panel to resolve a faulty fuel-injector tip; implemented supplier pre-ship containment that prevented the defect from escalating to a field issue or NHTSA recall.
- Improved model-based design process using MagicDraw to accelerate DFMEA development for engine systems.
- Mapped real-world customer usage and refined test methods, increasing in-field quality by **20%**.
- Redesigned operator instruction sheet after multiple plant Gemba visits, boosting UAW operator efficiency.

Purdue Electric Racing (Formula SAE EV)

Ex-Module Lead / Senior Battery Advisor

August 2023 - Present

- Designed a waterproof battery enclosure that achieved **100%** leak-free performance during tech inspection.
- Engineered a Simscape battery model to improve peak battery performance and increase pack efficiency.
- Led module redesign for space optimization, increasing energy density by **30%** in the new architecture.
- Designed a thermal interface and spacer system to improve cooling, enhancing battery performance by **10%**.
- Developed high-accuracy cooling plate validation test for edge cases using Arduino, helped achieve top 10 finish.

MathWorks Inc.

Campus Ambassador

January 2025 - January 2026

- Directed hands-on Simulink drone path workshop for student orgs, increasing student participation by **40%**.
- Organized technical workshops, student competitions, and department partnerships to increase Simulink adoption.
- Enabled students' access to MATLAB resources by hosting demos aligned with academic and research use cases.

Tredence Inc.

Data Science Intern

May 2024 - August 2024

- Identified sector trends using NumPy and Matplotlib libraries in Python to inform retail decisions for clients.
- Devised multiple regression models to analyze supply chain data, reducing costs for key customers by **15%**.
- Leveraged data cleaning, analysis, and visualization skills using SQL to drive scalable insights and strategies.

The Pushcart Project

Founder- Personal Project

January 2021 - August 2023

- Produced a functional electric waste-pushcart prototype in Bangalore, India, from the ground up for **<\$2000**.
- Donated prototype to a local waste market to assist elderly workers operate heavy carts, promoting social impact.
- Advocated for and facilitated transition from internal combustion alternatives to greener e-cart solutions.

Skills

C | MATLAB | DFMEA | SolidWorks | FEA | ROS2 | SQL | Python | GD&T | SPC | 3DX | DFM | Simulink | 3D printing